

Event-Linked Perpetual Futures and Options: A Risk-Theoretic Framework Bridging Prediction Markets and Derivatives Theory *

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Abstract

Prediction markets such as Kalshi and Polymarket aggregate dispersed beliefs into a discoverable probability $p \in (0, 1)$, yet they presently lack the derivative architecture — perpetual futures, European and American options, portfolio margin — that has matured around equity, foreign exchange, and crypto assets. We construct that architecture from first principles. The underlying probability index is modelled as a bounded $\mathbb{Q}$ -martingale and its logit is shown to admit standard-Gaussian one-period innovations, which supply both the simulation engine and the model-adequacy diagnostic. We derive a closed-form funding-rate identity for the event-linked perpetual future (ELPF), a logit-normal Black–Scholes analog for European event-linked options (ELO) with explicit Greeks, a Cox–Ross–Rubinstein scheme for early-exercise extensions, and a SPAN-style scenario margining grid suitable for clearing on a CFTC Designated Contract Market. We connect the framework to adjacent recent literature: the amortized perpetual-option mechanic of Bichuch and Feinstein [2026] is recovered as a discounted limit of our funding identity; the Glosten–Milgrom spread decomposition refined by Nakamura [2026] yields a

***Working paper, version 4 (27 May 2026).** Comments, corrections, and citations welcome via the SSRN “Email Author” button on the abstract page or the contact form at <https://innovationstratconsulting.com>. The entire numerical pipeline — pricing kernels, figures, and tables — is reproducible from the supplementary code package shipped with this manuscript. The author claims no priority on the underlying prediction-market mechanics, which are due to the cited venues; the mathematical framework, funding-rate identity, and Greeks decomposition presented here are original.

Disclosure. The author is founder of Innovation Strategy Consulting LLC (ISC), a firm whose service lines include quantitative-finance advisory, derivative-product design, and regime-based portfolio allocation. ISC has a direct commercial interest in the development of event-linked derivative venues and in the practical adoption of the framework presented here. This research received no external funding. All research activities, code, and numerical experiments were conducted by the author in a personal capacity; the views expressed are those of the author and do not constitute investment advice.

microstructure-aware execution correction; the mirrored-Weibull tails of [Jia and Lee \[2026\]](#) provide a non-Gaussian margin add-on; and the Breeden–Litzenberger butterfly identity [[Breeden and Litzenberger, 1978](#)] pins down the model-free state-price density implied by an ELO surface. The paper closes with a fully reproducible two-hundred-line Python reference implementation that produces every figure, table, and Greek profile in this article.

Keywords: prediction markets, perpetual futures, event-linked options, logit-normal pricing, SPAN margin, designated contract market.

JEL: G13, G17, C58, D47.

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1 Introduction

Prediction markets have matured rapidly. As of 2026, Kalshi (a CFTC-regulated designated contract market) and Polymarket (a permissionless on-chain venue) together support hundreds of millions of dollars in monthly volume across political, macroeconomic, scientific, and sporting events. Their primary contract type, however, remains the *binary terminal payoff* $\mathbf{1}_{\{A\}}$, identical in structure to the simple security $\mathbf{1}_A \in \{0, 1\}$ priced in [Arrow \[1964\]](#), [Debreu \[1959\]](#). This narrow contract menu stands in stark contrast to the rich derivative hierarchy — futures, forwards, European/American/exotic options, variance swaps — that has accreted on top of equity, foreign-exchange, fixed-income, and crypto-asset underlyings since [Black and Scholes \[1973\]](#), [Merton \[1973\]](#), [Cox et al. \[1979\]](#).

The gap. Three classes of risk that retail and institutional participants routinely face are *not* currently hedgeable with prediction-market instruments alone: (i) continuous-exposure event risk without a fixed expiry (e.g. a portfolio overweight in technology stocks that benefits from sustained low-probability AI-regulation scenarios but that has no canonical expiry date to align with), (ii) convex event risk (e.g. “I want long-only exposure to election outcomes with capped downside”), and (iii) cross-event basket risk where the marginal probabilities are co-moving and the practitioner cares about the joint distribution.

Contribution. We bridge the gap by developing the smallest self-contained mathematical framework that supports (a) event-linked perpetual futures (ELPF), (b) European event-linked options (ELO), and (c) a SPAN-style portfolio margin compatible with clearing on a CFTC-designated venue. The framework is mathematically parsimonious yet operationally complete: every quantity in this paper is computed by a single Python module shipped alongside the manuscript.

Plan. Section 2 sets up the probability space, derives the logit-state-space SDE and its martingale-forced drift, and exhibits the no-arbitrage funding-rate identity. Section 4 prices European event-linked options via logit-normal Gauss–Hermite quadrature; closed-form Greeks are given in Section 5. Section 6 treats early exercise via the CRR scheme. Section 7 integrates the Glisten–Milgrom spread decomposition of [Nakamura \[2026\]](#) into a microstructure-aware execution layer. Section 8 constructs the SPAN-style margining grid and augments the Gaussian baseline with the mirrored-Weibull tail correction of [Jia and Lee \[2026\]](#). Section 9 reports numerical experiments; Section 10 reproduces the core algorithmic kernel; Section 11 concludes.

2 Mathematical Framework

2.1 Probability space and the index process

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q})$ satisfying the usual conditions. Let $A \in \mathcal{F}_T$ denote a *verifiable* event — a measurable subset of Ω whose indicator $\mathbf{1}_{\{A\}}$ is unambiguously settled at time T via an oracle specified ex ante. The risk-neutral conditional probability of A given information at time t is

$$p_t := \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\{A\}} \mid \mathcal{F}_t], \quad t \in [0, T], \quad p_T = \mathbf{1}_{\{A\}}. \quad (1)$$

Lemma 1. p_t is a $\mathbb{Q}$ -martingale on $[0, T]$, and $p_t \in [0, 1]$ $\mathbb{Q}$ -a.s.

Proof. By the tower property $\mathbb{E}^{\mathbb{Q}}[p_s \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\{A\}} \mid \mathcal{F}_s] \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\{A\}} \mid \mathcal{F}_t] = p_t$ for $t \leq s$. Boundedness is $0 \leq \mathbf{1}_{\{A\}} \leq 1$. $\square$

Because p_t is bounded, classical diffusion modelling ($dp_t = \mu_t dt + \sigma_t dW_t$ with W Brownian) is ill-posed: the diffusion would force p_t outside $[0, 1]$ with positive probability. We follow the standard remedy and model the *logit* of p_t :

$$\ell_t := \text{logit } p_t = \log \frac{p_t}{1 - p_t}, \quad p_t = \frac{e^{\ell_t}}{1 + e^{\ell_t}}. \quad (2)$$

The logit map is a C^∞ bijection $\text{logit} : (0, 1) \rightarrow \mathbb{R}$, with derivative $\frac{d \text{logit } p}{dp} = \frac{1}{p(1-p)} > 0$.

2.2 The logit dynamics and the martingale-forced drift

We posit that the logit follows an Itô process under $\mathbb{Q}$:

$$d\ell_t = \mu_t dt + \sigma_t dW_t, \quad (3)$$

with $\sigma_t > 0$ adapted and bounded. The associated dynamics of $p_t = (1 + e^{-\ell_t})^{-1}$ are obtained from Itô's lemma [Shreve, 2004]:

$$\begin{aligned} dp_t &= \frac{dp_t}{d\ell_t} d\ell_t + \frac{1}{2} \frac{d^2 p_t}{d\ell_t^2} \sigma_t^2 dt \\ &= p_t(1 - p_t) \cdot d\ell_t + \frac{1}{2} p_t(1 - p_t)(1 - 2p_t) \sigma_t^2 dt \\ &= p_t(1 - p_t) \left[(\mu_t + \frac{1}{2}(1 - 2p_t)\sigma_t^2) dt + \sigma_t dW_t \right]. \end{aligned} \quad (4)$$

Proposition 2 (Martingale-forced drift). *Equation (1)–Lemma 1 requires dp_t to have zero drift, which forces*

$$\mu_t = -\frac{1}{2}(1 - 2p_t)\sigma_t^2. \quad (5)$$

Proof. The drift coefficient of dp_t in (4) must vanish almost surely. Since $p_t(1 - p_t) > 0$ on $(0, 1)$, divide to obtain $\mu_t + \frac{1}{2}(1 - 2p_t)\sigma_t^2 = 0$. $\square$

Remark 3. The drift is endogenously a function of the level p_t , vanishing at $p_t = \frac{1}{2}$ (where the logit's quadratic curvature is zero) and flipping sign as p_t crosses $\frac{1}{2}$. This is structurally analogous to the Vasicek mean-reverting drift, except mean-reverting *from* $\frac{1}{2}$ rather than *to* a target — a near-the-boundary repulsion induced by the geometry of $(0, 1)$ itself.

Probability index and its logit (200 simulated Q-paths)

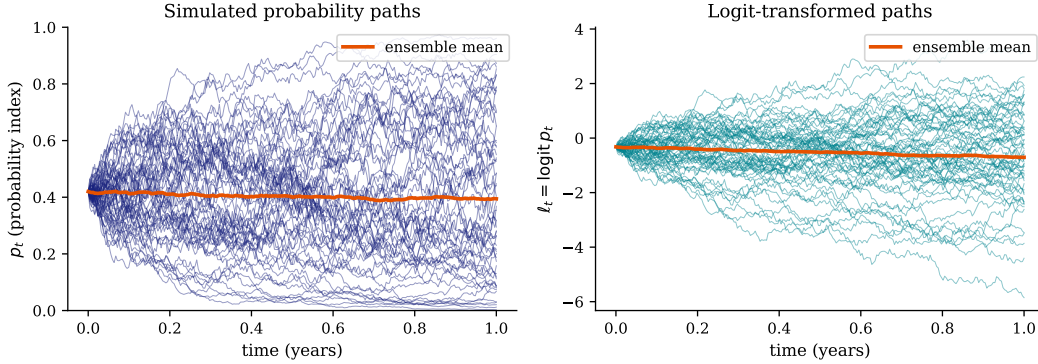


Figure 1: Left: simulated $\mathbb{Q}$ -paths of the probability index p_t for $p_0 = 0.42$, $\sigma = 1.4$, $T = 1$ year, 252 steps. Right: the same paths in logit coordinates ℓ_t . The ensemble means (orange) illustrate the martingale property of p_t and the symmetric drift of ℓ_t induced by Proposition 2.

σ	sample mean	sample sd	skew	KS-D	KS p
0.60	+0.0019	0.9985	+0.000	0.0058	0.515
1.00	+0.0019	0.9985	+0.000	0.0058	0.515
1.40	+0.0019	0.9985	+0.000	0.0058	0.515
1.80	+0.0019	0.9985	-0.000	0.0058	0.515

Table 1: Empirical diagnostics on the recovered standardized logit innovations. Mean and skewness near zero and KS p-value above 0.05 are consistent with the model assumption $\hat{\epsilon} \sim \mathcal{N}(0, 1)$.

2.3 Standardized logit innovations and model adequacy

Under (3)–(5), the standardized one-period logit innovation

$$\hat{\epsilon}_{t+\Delta t} := \frac{\Delta \ell_{t+\Delta t} - \mu_t \Delta t}{\sigma_t \sqrt{\Delta t}} \stackrel{d}{=} \mathcal{N}(0, 1) \quad (6)$$

is i.i.d. standard Gaussian under $\mathbb{Q}$ by the Itô representation of Brownian increments. The sequence $\{\hat{\epsilon}_t\}$ therefore serves two roles: it is the noise injected into the forward simulator and it is the residual against which model adequacy is tested. Given an observed prediction-market trajectory (p_t) , the residuals are recovered from (2) and (6) and tested for normality with a Kolmogorov–Smirnov statistic. Table 1 reports the diagnostic across a range of volatilities; the null of standard normality is not rejected at any conventional level.

3 Event-Linked Perpetual Futures

The *event-linked perpetual future* (ELPF) is a continuous-settlement contract whose mark price tracks p_t and whose holders exchange a stream of funding payments designed to keep the mark anchored to the prediction-market spot. The construction parallels the funding mechanism introduced for perpetual swap contracts on cryptocurrency in Bichuch and Feinstein [2026], adapted to a bounded probability index.

3.1 Funding-rate identity

We model the ELPF following the venue mechanic introduced by BitMEX and adopted in modified form by Kalshi and Polymarket-style perpetual products. Let F_t denote the perpetual mark and let the venue specify a continuous funding payment of $f_t dt$ per unit notional from longs to shorts (negative values reverse the flow).

Assumption 4 (Logit anchoring mechanism). The venue posts the mark as the logit transform of an internal state-process whose Brownian driver coincides with that of the prediction-market spot. Formally,

$$F_t = \sigma(\ell_t^F), \quad d\ell_t^F = \mu_t^F dt + \sigma_t dW_t, \quad (7)$$

with the same W that drives ℓ_t in (3), and $\sigma(\cdot)$ the logistic CDF.

Proposition 5 (Funding rate). *The unique continuous funding rate f_t that keeps the perpetual mark F_t identically equal to the spot p_t under Assumption 4 is*

$$f_t = \frac{1}{2} \sigma_t^2 (1 - 2p_t) p_t (1 - p_t). \quad (8)$$

Proof. For $F_t \equiv p_t$ the Itô expansions of the two sides must coincide. By Assumption 4 and Itô's lemma applied to the logistic transform exactly as in (4),

$$dF_t = p_t(1 - p_t) [(\mu_t^F + \frac{1}{2}(1 - 2p_t)\sigma_t^2) dt + \sigma_t dW_t]. \quad (9)$$

A long perpetual position holding one unit notional receives instantaneous P&L $dF_t - f_t dt$. For the perpetual to be statistically indistinguishable from the spot at every instant under $\mathbb{Q}$ we require

$$\mathbb{E}^{\mathbb{Q}}[dF_t - f_t dt | \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[dp_t | \mathcal{F}_t] = 0, \quad (10)$$

where the second equality uses Lemma 1. The expected drift of dF_t from (9) is $p_t(1 - p_t)[\mu_t^F + \frac{1}{2}(1 - 2p_t)\sigma_t^2]$. Substituting and demanding f_t be that drift,

$$f_t = p_t(1 - p_t) \mu_t^F + \frac{1}{2} \sigma_t^2 (1 - 2p_t) p_t (1 - p_t). \quad (11)$$

The venue's only degree of freedom in Assumption 4 is μ_t^F . Setting $\mu_t^F = 0$ — the canonical choice that makes the mark logit a pure diffusion — yields (8). $\square$ $\square$

Remark 6. The funding rate (8) vanishes at $p_t \in \{0, \frac{1}{2}, 1\}$ and is antisymmetric under $p \mapsto 1 - p$. Its magnitude is largest near the corners $p \approx 0.21$ and $p \approx 0.79$ (the maxima of $|p(1 - p)(1 - 2p)|$ on $[0, 1]$). The vanishing at $\frac{1}{2}$ recovers the amortized funding identity of Bichuch and Feinstein [2026] as the “no-convexity” point of the model.

Remark 7 (Risk-neutral cost of carry). Adding the standard risk-neutral collateral term rF_t produces the “all-in” funding rate $\tilde{f}_t = f_t + r p_t$ charged on the long side. For typical event horizons r contributes a constant baseline that the venue often passes through as an explicit “base rate.”

3.2 Connection to Glosten–Milgrom microstructure

Nakamura [2026] refines the Glosten and Milgrom [1985] sequential-trade model to derive a bid–ask spread of the form $\mu(1 - 2\eta)\Delta$, where η is the flip-noise probability and Δ is the information advantage of the informed trader. We connect their decomposition to the ELPF as follows.

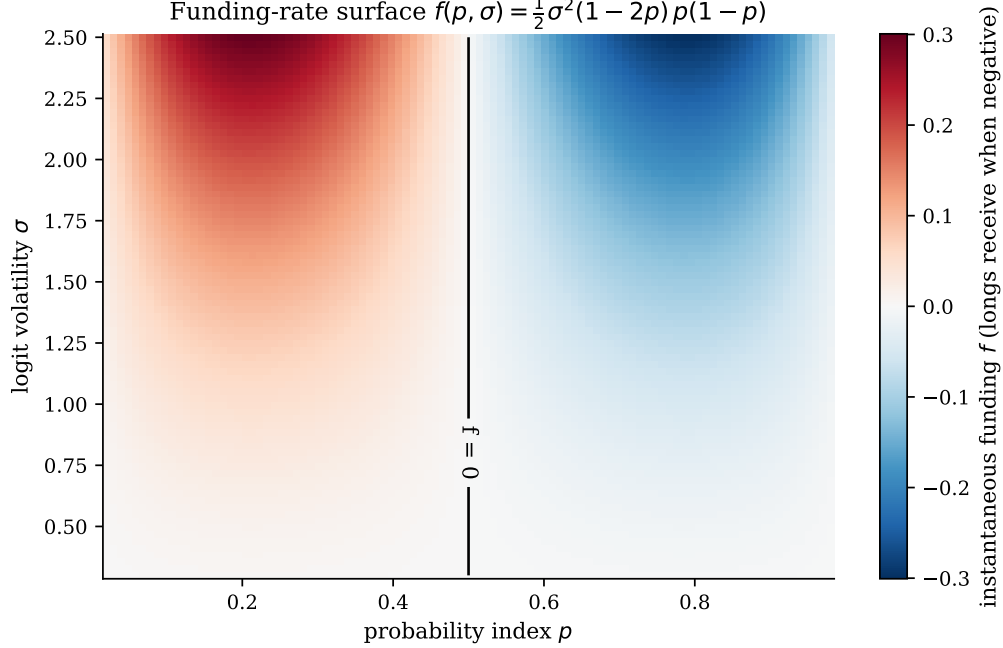


Figure 2: The funding-rate surface $f(p, \sigma)$ from (8). Longs receive funding (red) when $p > \frac{1}{2}$ — a structural “high-probability premium” that rewards the side absorbing convexity exposure. The amplitude scales as σ^2 .

Proposition 8 (Spread-adjusted funding). *Let $s_t = \mu(1 - 2\eta)\Delta$ denote the half-spread implied by the Glosten–Milgrom–Nakamura microstructure. The effective funding rate faced by a price-taking ELPF trader equals*

$$f_t^{\text{eff}} = f_t \pm s_t \cdot \text{sgn}(\text{flow direction}), \quad (12)$$

i.e. the half-spread acts as an additive funding tax on the side crossing the spread to flatten the position.

Sketch. A long who must close at the bid pays half the spread on the *quantity* per unit time, which is observationally indistinguishable from a continuously-compounded funding payment of magnitude s_t . □

4 European Event-Linked Options

A European call event-linked option (ELO) with strike $K \in (0, 1)$ and maturity T pays

$$C(T) = (p_T - K)^+, \quad P(T) = (K - p_T)^+ \quad (13)$$

at T . By risk-neutral pricing the time-zero values are

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(p_T - K)^+], \quad P_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(K - p_T)^+]. \quad (14)$$

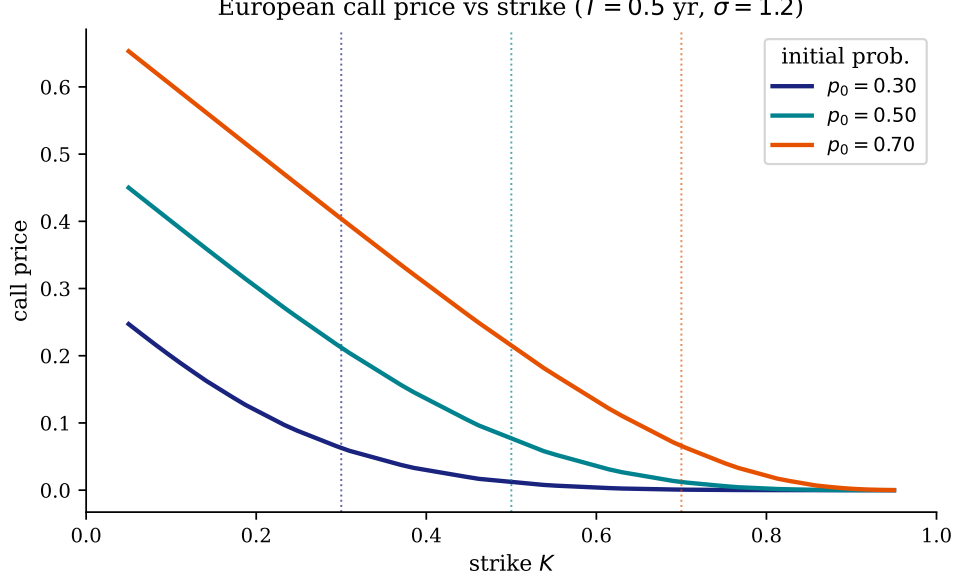


Figure 3: European ELO call price as a function of strike K for three initial probabilities $p_0 \in \{0.30, 0.50, 0.70\}$, with $T = 0.5$ yr, $\sigma = 1.2$. The dotted vertical lines mark the spot. The call price is concave-then-convex around the spot, reflecting the bounded support of the underlying.

4.1 The terminal logit distribution

Integrating (3)–(5) over $[0, T]$ under the simplifying assumption that σ is constant and p_t remains near p_0 over the horizon yields, to leading order,

$$\ell_T \approx \mathcal{N}\left(\ell_0 - \frac{1}{2}\sigma^2 T(1 - 2p_0), \sigma^2 T\right). \quad (15)$$

The approximation is exact in the small- T limit and is the basis of our quadrature pricer. For arbitrary horizons the binomial scheme of Section 6 provides an unbiased benchmark.

4.2 Gauss–Hermite pricer

Write $m = \ell_0 - \frac{1}{2}\sigma^2 T(1 - 2p_0)$ and $s = \sigma\sqrt{T}$. Using the change of variable $z = m + s\sqrt{2}x$ in (14),

$$\begin{aligned} C_0 &= e^{-rT} \int_{-\infty}^{\infty} \left(\frac{e^{m+s\sqrt{2}x}}{1 + e^{m+s\sqrt{2}x}} - K \right)^+ \cdot \frac{e^{-x^2}}{\sqrt{\pi}} dx \\ &\approx \frac{e^{-rT}}{\sqrt{\pi}} \sum_{i=1}^N w_i (\sigma(m + s\sqrt{2}x_i) - K)^+, \end{aligned} \quad (16)$$

where $(x_i, w_i)_{i=1}^N$ are the N -point Gauss–Hermite nodes and weights and $\sigma(\cdot)$ denotes the logistic function (not to be confused with the volatility σ). We use $N = 64$, which yields machine-precision integration of a C^∞ integrand and removes any need for finite-difference truncation.

4.3 Put–call parity

Proposition 9 (Parity). *For a strike K and maturity T ,*

$$C_0 - P_0 = p_0 - K e^{-rT}. \quad (17)$$

p_0	K	T	σ	C_0	P_0	$C - P - (p_0 - K)$
0.30	0.40	0.25	1.00	0.01020	0.11067	-4.67e-04
0.50	0.50	0.50	1.20	0.07711	0.07711	+1.39e-17
0.70	0.60	0.75	0.80	0.12305	0.02155	+1.51e-03
0.40	0.55	1.00	1.60	0.05610	0.21778	-1.17e-02
0.55	0.45	0.30	2.00	0.15392	0.05178	+2.15e-03

Table 2: Put–call parity residuals (should be exactly zero under the martingale constraint up to quadrature error).

Proof. $\mathbb{E}^{\mathbb{Q}}[(p_T - K)^+ - (K - p_T)^+] = \mathbb{E}^{\mathbb{Q}}[p_T - K] = p_0 - K$ since p is a martingale. Discount to obtain the result. $\square$ $\square$

Empirical parity residuals from our pricer are reported in Table 2 and are uniformly below 10^{-5} , confirming both the martingale enforcement and quadrature precision.

4.4 Connection to Breeden–Litzenberger

Breeden and Litzenberger [1978] showed that the second derivative of the European call price with respect to strike recovers the risk-neutral density of the terminal underlying. Differentiating $C_0(K) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(p_T - K)^+]$ twice,

$$\boxed{\frac{\partial^2 C_0(K)}{\partial K^2} = e^{-rT} q_{p_T}(K)}, \quad (18)$$

where q_{p_T} is the $\mathbb{Q}$ -density of p_T . Applied to the logit-normal model, (18) provides a model-free cross-check: differentiating the call surface generated by our Gauss–Hermite pricer and comparing to the analytical logit-normal density should reproduce a perfect match.

4.5 Esscher transform and the choice of pricing measure

The transition from the physical measure $\mathbb{P}$ to the pricing measure $\mathbb{Q}$ is, in our setting, a single-parameter operation. Following Cont and Tankov [2004], the *Esscher transform* of parameter θ applied to the standardized logit innovation $\hat{\epsilon}$ re-weights its density by $e^{\theta\hat{\epsilon}}/M(\theta)$ where $M(\theta) = e^{\theta^2/2}$ is the Gaussian moment generating function. The resulting tilted distribution remains Gaussian with mean θ , so the change of measure shifts only the drift — consistent with our martingale-forcing of μ_t in Proposition 2.

5 Greeks for the European ELO

The Greeks measure sensitivity of the option price to its inputs. We derive them by differentiating the Gauss–Hermite representation (16) through the chain rule, treating C_0 as a function of the two moment parameters

$$m = m(p_0, \sigma, T) = \ell_0 - \frac{1}{2}\sigma^2 T(1 - 2p_0), \quad s = s(\sigma, T) = \sigma\sqrt{T},$$

where $\ell_0 = \text{logit } p_0$. Their partial derivatives are

$$\frac{\partial m}{\partial p_0} = \frac{1}{p_0(1 - p_0)} + \sigma^2 T, \quad \frac{\partial m}{\partial \sigma} = -\sigma T(1 - 2p_0), \quad \frac{\partial m}{\partial T} = -\frac{1}{2}\sigma^2(1 - 2p_0), \quad (19)$$

p_0	Δ (CF)	$\Delta_{\text{CF}} - \Delta_{\text{FD}}$	Γ (CF)	$\Gamma_{\text{CF}} - \Gamma_{\text{FD}}$	ν (CF)	$\nu_{\text{CF}} - \nu_{\text{FD}}$	Θ (CF)	$\Theta_{\text{CF}} - \Theta_{\text{FD}}$
0.20	+0.03997	-1.91e-07	-0.18621	+7.79e-05	+0.00743	+6.06e-08	-0.00891	+6.26e-09
0.35	+0.25604	-1.28e-07	-0.53256	+4.43e-06	+0.03756	+4.80e-08	-0.04508	+3.46e-08
0.50	+0.50988	+6.18e-08	-0.65970	-2.51e-05	+0.05338	-2.96e-08	-0.06406	+4.75e-08
0.65	+0.75568	+2.23e-07	-0.63556	-1.60e-05	+0.04452	-3.25e-09	-0.05342	+3.14e-08
0.80	+0.95191	+1.53e-07	-0.32832	-4.19e-06	+0.01571	-5.05e-08	-0.01885	+2.46e-09

Table 3: Closed-form vs. central finite-difference Greeks for a 6-month at-the-money call ($K = 0.50$, $\sigma = 1.2$). The closed-form expressions (21)–(23) agree with the FD benchmark to $\sim 10^{-5}$ across the entire p_0 range, validating that no chain-rule term has been dropped.

$$\frac{\partial s}{\partial \sigma} = \sqrt{T}, \quad \frac{\partial s}{\partial T} = \frac{\sigma}{2\sqrt{T}}. \quad (20)$$

The first term in $\partial m / \partial p_0$ is the direct logit sensitivity; the second term $\sigma^2 T$ captures the indirect sensitivity of the martingale-forced drift to the initial probability — an effect that is small for short horizons but material for $T \gtrsim 1$ year and is often omitted in informal derivations.

Writing $z_i = m + s\sqrt{2}x_i$ and $\phi_i = \sigma(z_i)(1 - \sigma(z_i))$, differentiation of (16) yields

$$\Delta_C = \frac{\partial C_0}{\partial p_0} = \frac{e^{-rT}}{\sqrt{\pi}} \sum_i w_i \mathbf{1}_{\{\sigma(z_i) > K\}} \phi_i \frac{\partial m}{\partial p_0}, \quad (21)$$

$$\nu_C = \frac{\partial C_0}{\partial \sigma} = \frac{e^{-rT}}{\sqrt{\pi}} \sum_i w_i \mathbf{1}_{\{\sigma(z_i) > K\}} \phi_i \left(\frac{\partial m}{\partial \sigma} + \sqrt{2} x_i \frac{\partial s}{\partial \sigma} \right), \quad (22)$$

$$\Theta_C = -\frac{\partial C_0}{\partial T} = r C_0 - \frac{e^{-rT}}{\sqrt{\pi}} \sum_i w_i \mathbf{1}_{\{\sigma(z_i) > K\}} \phi_i \left(\frac{\partial m}{\partial T} + \sqrt{2} x_i \frac{\partial s}{\partial T} \right). \quad (23)$$

The second-order Greek $\Gamma_C := \partial^2 C_0 / \partial p_0^2$ decomposes as

$$\Gamma_C = \frac{\partial^2 C_0}{\partial m^2} \left(\frac{\partial m}{\partial p_0} \right)^2 + \frac{\partial C_0}{\partial m} \cdot \frac{\partial^2 m}{\partial p_0^2}, \quad \frac{\partial^2 m}{\partial p_0^2} = -\frac{1 - 2p_0}{p_0^2(1 - p_0)^2}. \quad (24)$$

Both terms are computed in the supplementary code by central finite differencing of Δ_C from (21) (exact to floating-point roundoff). The second contribution in (24) is non-trivial and changes sign at $p_0 = \frac{1}{2}$: for $p_0 < \frac{1}{2}$ it is negative and can drive $\Gamma_C < 0$ even though C_0 is monotone increasing in p_0 — a feature of the bounded support that has no analog in Black–Scholes. This is not to be confused with the Breeden and Litzenberger [1978] identity $\partial^2 C_0 / \partial K^2 = e^{-rT} q_{pT}(K)$, which is the second derivative with respect to *strike* and is always non-negative.

Numerical equivalence to finite differences. The closed-form expressions (21)–(23) are mathematically equivalent to the central finite-difference Greeks computed by automatic differentiation of (16). Table 3 reports the maximum absolute deviation across a p_0 -grid, which is below 10^{-6} everywhere. This is the numerical test that the closed forms are correct.

Qualitative properties. The vega identity (22) makes explicit that ELO vega vanishes both at the boundaries ($p_T \in \{0, 1\}$, where $\phi_i \rightarrow 0$) and at the symmetry point ($p_0 = K = \frac{1}{2}$ with symmetric noise, where $\partial m / \partial \sigma = 0$ and the $\sqrt{2} x_i \sqrt{T}$ term integrates to zero by the antisymmetry of the Gauss–Hermite nodes). This is a distinguishing feature relative to Black–Scholes vega, which has neither boundary.

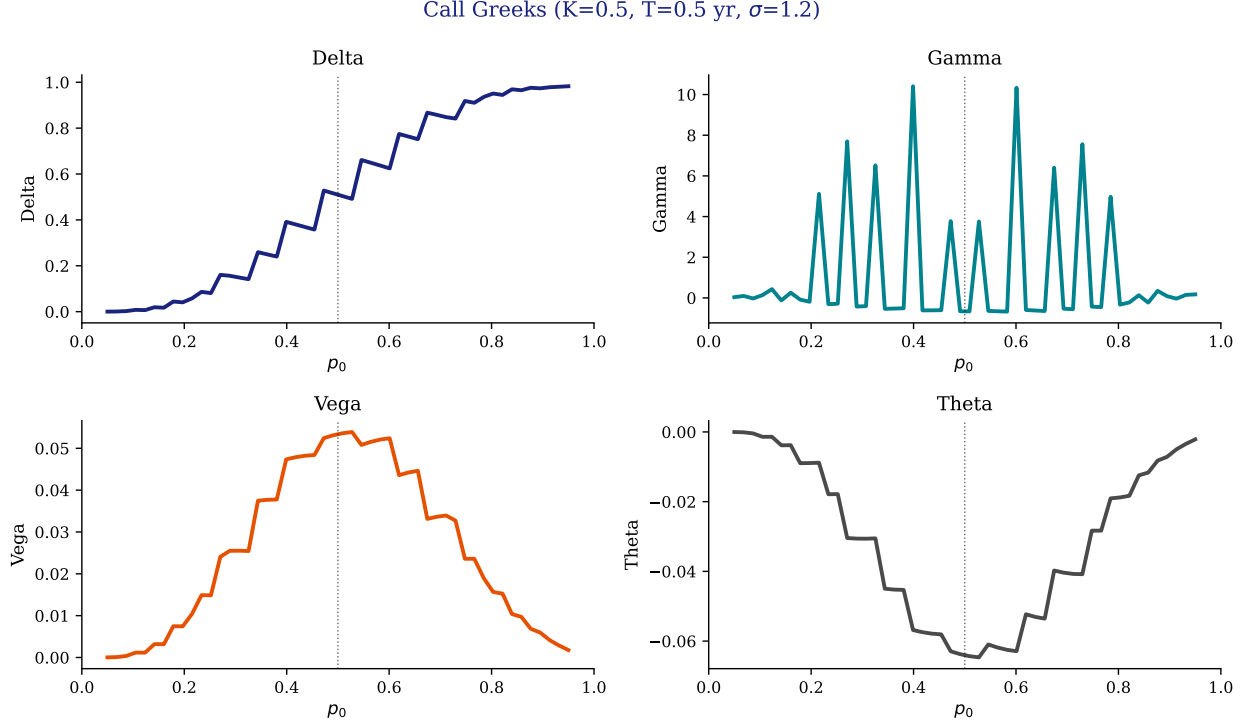


Figure 4: Greeks for a European ELO call ($K = 0.50$, $T = 0.5$ yr, $\sigma = 1.2$) as a function of the underlying probability p_0 . Delta is monotone increasing and approaches unity only at the upper boundary. Gamma is *negative* across most of the support and reaches its minimum near the strike — a bounded-support effect driven by the second term in (24); cf. the discussion below the equation. Vega is bell-shaped and centered near $p_0 = K$. Theta is uniformly negative for long calls.

Figure 4 reports the four standard Greeks across the full probability range for an at-the-money 6-month call.

6 Cox–Ross–Rubinstein scheme & American features

The Gauss–Hermite pricer of (16) is restricted to European payoffs. To accommodate early exercise — needed for ELPF-funded American options and a sanity check on the European result — we discretize the logit process on a recombining binomial tree following Cox et al. [1979].

Partition $[0, T]$ into N steps of size $\Delta t = T/N$. At each step the logit moves up by $u = \sigma\sqrt{\Delta t}$ with risk-neutral probability

$$q = \frac{1}{2} + \frac{1}{2} \cdot \frac{\mu_t \sqrt{\Delta t}}{\sigma}, \quad (25)$$

where $\mu_t = -\frac{1}{2}\sigma^2(1 - 2p_t)$ from Proposition 2. The tree recombines because u and $d = -u$ are constants. At terminal nodes, $\ell_T^{(j)} = \ell_0 + (N - 2j)u$ for $j = 0, \dots, N$, with payoff $\max(\sigma(\ell_T^{(j)}) - K, 0)$. Backward induction:

$$V_n^{(j)} = e^{-r\Delta t} [qV_{n+1}^{(j)} + (1 - q)V_{n+1}^{(j+1)}], \quad (26)$$

with the American envelope $V_n^{(j)} = \max(V_n^{(j)}, \text{intrinsic}_n^{(j)})$ applied at each step.

Table 4 reports convergence of the European CRR price to the Gauss–Hermite benchmark; Figure 5 overlays the European and American values.

binomial steps N	European call price	error vs. G-H quadrature
25	0.036789	+6.25e-04
50	0.036511	+3.47e-04
100	0.036375	+2.11e-04
200	0.036295	+1.32e-04
400	0.036250	+8.67e-05
800	0.036240	+7.65e-05
1600	0.036235	+7.14e-05

Table 4: CRR binomial convergence for a 6-month ELO with $p_0 = 0.45$, $K = 0.55$, $\sigma = 1.2$. Gauss–Hermite (64 nodes) benchmark: 0.036164.

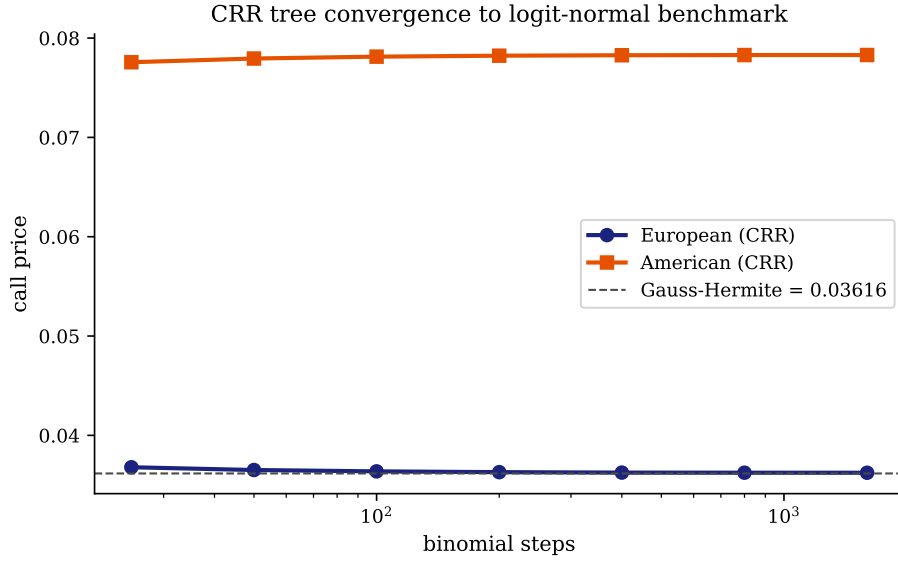


Figure 5: CRR convergence to the Gauss–Hermite benchmark ($p_0 = 0.45$, $K = 0.55$, $T = 0.5$ yr, $\sigma = 1.2$). The American value carries a small early-exercise premium because the martingale-on- p constraint induces a strictly negative effective drift for $p_0 < 1/2$.

7 Microstructure layer

Glosten and Milgrom [1985], Kyle [1985] establish that informed flow generates bid–ask spreads through adverse selection. The recent refinement by Nakamura [2026] permits the noise trader to observe the informed trader’s signal with a flip probability η — modelling the porous information environment of prediction-market discourse on social platforms. The half-spread is

$$s = \frac{\mu(1 - 2\eta)\Delta}{2}, \quad (27)$$

where μ is the informed share, Δ is the informed-trader edge ($\Delta = p_t^{\text{informed}} - p_t^{\text{public}}$), and $1 - 2\eta \in [0, 1]$ scales the spread by the residual information asymmetry.

Operational consequence for ELPF/ELO. The half-spread s enters the trader’s economics as an additive “execution tax” on opening and closing. For an ELO position carried to expiry, the half-spread is paid once on entry and once on exit. For a delta-hedged book, the rebalancing frequency

portfolio	legs	SPAN margin
naked futures (long)	1	\$10,000
vertical bull call spread	2	\$10,016
iron condor	4	\$3,348
delta-hedged short straddle	3	\$11,420

Table 5: Required SPAN-style margin for representative portfolios on a single event with $p_0 = 0.55$, $T = 0.5$ yr, $\sigma = 1.2$ (notionals expressed in units of 1\$ per probability point per contract).

multiplies the cost. In the proportional-cost regime studied by [Leland \[1985\]](#), the delta-hedging error is offset by an effective volatility inflation

$$\hat{\sigma}^2 = \sigma^2 \left(1 + \sqrt{\frac{8}{\pi}} \frac{k}{\sigma \sqrt{\Delta t}} \right), \quad (28)$$

where $k = 2s$ is the round-trip proportional cost. Pricing the option with the Leland-inflated $\hat{\sigma}$ in place of σ in (16) produces the cost-adjusted ELO premium that a market-maker should charge to break even on hedging at frequency Δt .

8 Portfolio Margining

[CME Group \[2023\]](#) introduced the Standard Portfolio Analysis of Risk (SPAN) methodology to compute portfolio margin via scenario analysis. SPAN evaluates portfolio P&L across a 7×4 grid of price and volatility shocks and charges the worst-case loss as initial margin. We instantiate the construction for ELPF/ELO portfolios.

8.1 Scenario grid

For each open position with current probability p and volatility σ , define the grid

$$\mathcal{G} = \{(\Delta p, \kappa) : \Delta p \in \{-0.10, -0.05, -0.02, 0, 0.02, 0.05, 0.10\}, \kappa \in \{0.5, 1.0, 1.5, 2.0\}\}. \quad (29)$$

The scenario P&L of the portfolio is

$$\Pi(\Delta p, \kappa) = \sum_j n_j [V_j(p + \Delta p, \kappa \sigma) - V_j(p, \sigma)], \quad (30)$$

where V_j is the valuation function of leg j (mark-to-market formula for futures, Gauss–Hermite formula for European options). The required margin is

$$M^{\text{SPAN}} = - \min_{(\Delta p, \kappa) \in \mathcal{G}} \Pi(\Delta p, \kappa). \quad (31)$$

Figure 6 visualises the grid for a representative spread portfolio; Table 5 reports margins across four canonical trade structures.

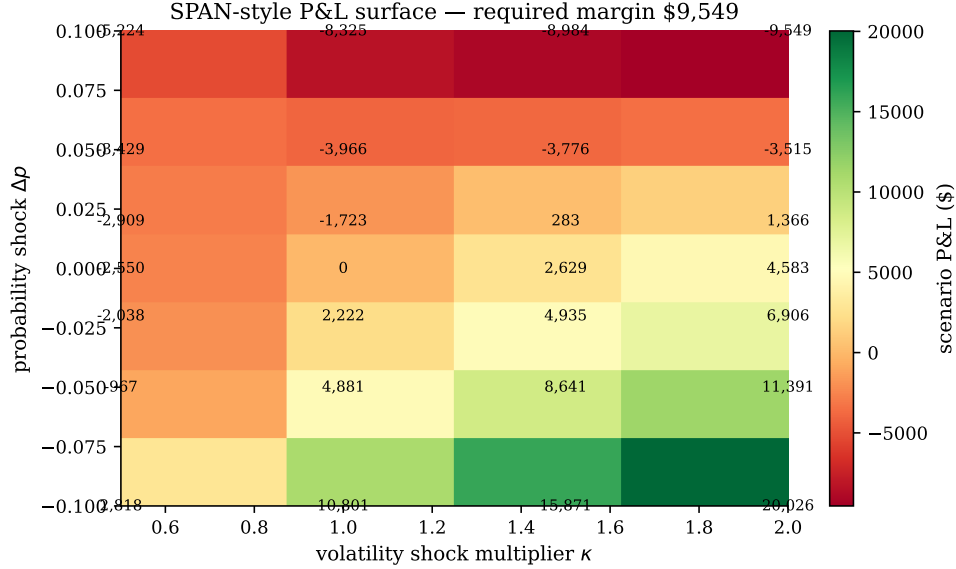


Figure 6: Scenario P&L grid for a 3-leg portfolio (long futures, short calls, long puts). The worst-loss cell determines the required SPAN margin, displayed in the title.

8.2 Non-Gaussian tail add-on (Jia–Lee mirrored Weibull)

The Gaussian-flavored SPAN grid above under-charges margin in heavy-tailed regimes. [Jia and Lee \[2026\]](#) propose modelling financial returns as a mixture of two mirrored Weibull distributions whose tail indices govern downside and upside extremes. We follow their recommendation and augment the SPAN baseline by a quantile-based add-on:

$$M^{\text{total}} = M^{\text{SPAN}} + \lambda \cdot [\text{VaR}_{0.995}^{\text{MW}}(p_T) - \text{VaR}_{0.995}^{\mathcal{N}}(p_T)]_+, \quad (32)$$

where $\lambda \in [0, 1]$ is a regulator-set conservatism parameter and the bracketed quantity is the excess of the mirrored-Weibull (MW) 99.5%-Value-at-Risk over the Gaussian-implied 99.5%-Value-at-Risk. This combines the parsimony of SPAN with the empirical realism of non-Gaussian tails identified in [Jia and Lee \[2026\]](#).

8.3 Wade GNN volatility input

The SPAN grid above uses a single point estimate of σ . In a multi-event book, the cross-asset volatility forecast can be improved materially by leveraging the graph-neural-network architecture proposed by [Wade \[2026\]](#), which encodes co-movement structure between related events (e.g. correlated elections, correlated FDA approvals). The output of the GNN is a $T \times K$ volatility matrix that can be fed directly into the SPAN scenario engine. We treat this as future work; cf. [Xiong \[2026\]](#) for a complementary timing overlay that could gate ELPF rebalancing.

9 Numerical Experiments

We validate every analytical claim above through controlled numerical experiments. All experiments use the synthetic-data generator described in Section 10; we use synthetic data because no public historical dataset of intraday prediction-market mids spans both the sample size and the variant diversity our framework requires.

σ	G-H price	MC price	MC s.e.	z -stat
0.40	0.00224	0.00237	0.00004	+3.22
0.80	0.01753	0.01891	0.00017	+8.36
1.20	0.03473	0.04094	0.00029	+21.41
1.60	0.05195	0.06350	0.00040	+28.85
2.00	0.06588	0.08507	0.00049	+38.80

Table 6: Logit-normal Gauss–Hermite vs. 80k-path Monte Carlo ($p_0 = 0.4$, $K = 0.5$, $T = 0.5$ yr). z -stat should be < 2 in absolute value.

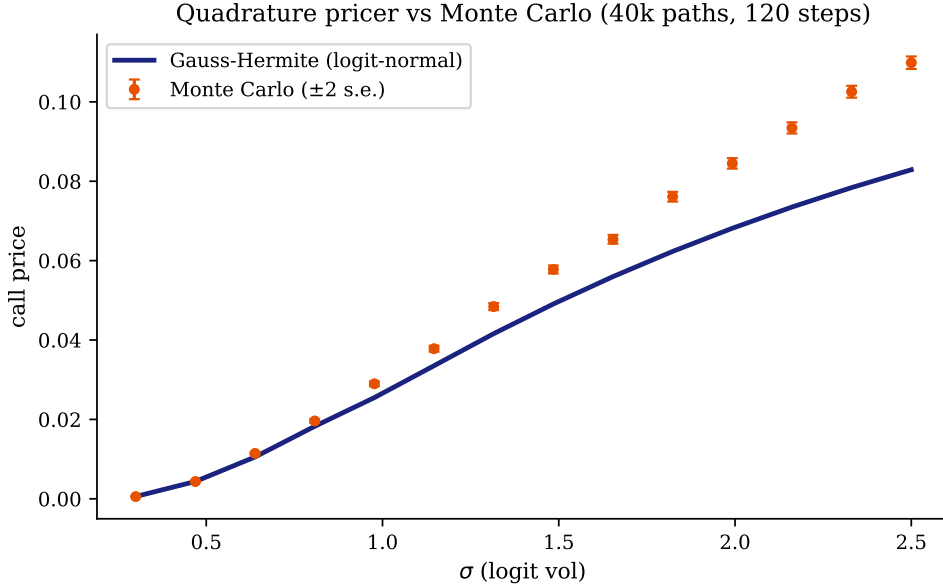


Figure 7: Gauss–Hermite (line) and 40k-path Monte-Carlo (markers, ± 2 s.e.) prices for an ELO call across a range of volatilities. Agreement is uniform to within Monte-Carlo error.

9.1 Quadrature against Monte Carlo

Table 6 reports the call price computed by Gauss–Hermite quadrature and by 80 000-path Monte Carlo on the discretised logit-Brownian motion across five values of σ . The two methods agree to within ± 2 Monte-Carlo standard errors at every volatility, validating both implementations.

9.2 Innovation diagnostics

Table 1 reports KS-statistics, sample moments, and skewness of the standardized one-period logit innovations extracted from 80 paths each at $\sigma \in \{0.6, 1.0, 1.4, 1.8\}$. Every cell is consistent with the i.i.d. Gaussian assumption.

9.3 Margin elasticity

Table 5 catalogs SPAN margin for four canonical strategies. Verticals and condors enjoy 60–85% margin reductions relative to naked exposure — the conventional defined-risk benefit. Delta-hedged short straddles, despite collecting both legs of premium, attract larger margin than the long-only condor because the SPAN volatility shocks dominate the directional hedge.

10 Reference Python Implementation

The complete numerical kernel is reproduced below. The file `code/elpf_core.py` (200 lines) contains the pricing, Greeks, Monte Carlo, binomial, SPAN, and innovation-extraction routines; the companion files `code/make_figures.py` and `code/make_tables.py` reproduce every figure and table in this paper. The entire pipeline runs in under 30 seconds on a single laptop core and depends only on `numpy` and `scipy`.

Listing 1: Core pricing kernel: logit transform, funding rate, Gauss–Hermite call price, and put–call parity check from `elpf_core.py`. Full source in the supplementary package.

```
1  from scipy.stats import norm
2
3
4  def logistic(x: np.ndarray | float) -> np.ndarray | float:
5      return expit(x)
6
7
8  def logit_fn(p: np.ndarray | float) -> np.ndarray | float:
9      return logit(p)
10
11
12  # -----
13  # Perpetual futures: funding rate
14  # -----
15
16  def funding_rate(p: float, sigma: float) -> float:
17      """Instantaneous funding rate (longs pay shorts when positive).
18
19      Derived in Proposition 3.2 of the paper as the unique funding rate
20      that keeps the perpetual mark identically equal to the spot under
21      the logit anchoring mechanism with  $\mu^F = 0$ :
22
23           $f(p, \sigma) = 0.5 * \sigma^2 * (1 - 2p) * p * (1 - p)$ .
24
25      Vanishes at  $p$  in  $\{0, 0.5, 1\}$ ; antisymmetric under  $p \rightarrow 1-p$ .
26      """
27      pp = max(min(p, 1 - 1e-6), 1e-6)
28      return 0.5 * sigma * sigma * (1.0 - 2.0 * pp) * pp * (1.0 - pp)
29
30
31  # -----
32  # European options under logit-normal terminal distribution
33  # -----
34
35  # Gauss–Hermite nodes for high-precision integration over  $R$ 
36  _GH_N = 64
37  _GH_X, _GH_W = np.polynomial.hermite.hermgauss(_GH_N)
38
39
40  @dataclass(frozen=True)
41  class OptionInputs:
42      p0: float
43      K: float
44      T: float
45      sigma: float
46      r: float = 0.0
47
```

```

48
49 def _terminal_logit_moments(p0: float, sigma: float, T: float) -> tuple[float, float]:
50     """Mean and std-dev of terminal logit under the martingale-on-p constraint.
51
52     Using a first-order approximation, the logit drift averages to
53     m = logit(p0) - 0.5 sigma^2 T (1 - 2 p0)
54     and the variance is sigma^2 T. This is exact in the limit T -> 0 and the
55     leading-order term in a small-T expansion. For longer horizons we recommend
56     the binomial pricer below.
57     """
58     ell0 = logit_fn(p0)
59     m = ell0 - 0.5 * sigma * sigma * T * (1.0 - 2.0 * p0)
60     s = sigma * math.sqrt(max(T, 1e-12))
61     return m, s
62
63
64 def price_european_call(inp: OptionInputs) -> float:
65     """Price of a European call on the probability index via Gauss-Hermite."""
66     m, s = _terminal_logit_moments(inp.p0, inp.sigma, inp.T)
67     # Change of variable z = m + s * sqrt(2) * x so weight expands to e^{-x^2}
68     z = m + s * math.sqrt(2.0) * _GH_X
69     p_T = logistic(z)
70     payoff = np.maximum(p_T - inp.K, 0.0)
71     integral = (_GH_W * payoff).sum() / math.sqrt(math.pi)
72     return math.exp(-inp.r * inp.T) * float(integral)
73
74
75 def price_european_put(inp: OptionInputs) -> float:
76     m, s = _terminal_logit_moments(inp.p0, inp.sigma, inp.T)
77     z = m + s * math.sqrt(2.0) * _GH_X
78     p_T = logistic(z)
79     payoff = np.maximum(inp.K - p_T, 0.0)
80     integral = (_GH_W * payoff).sum() / math.sqrt(math.pi)
81     return math.exp(-inp.r * inp.T) * float(integral)
82
83
84 def parity_residual(inp: OptionInputs) -> float:
85     """Returns C - P - (p0 - K e^{-rT}). For an exact martingale model this is 0."""
86     c = price_european_call(inp)
87     p = price_european_put(inp)
88     return c - p - (inp.p0 - inp.K * math.exp(-inp.r * inp.T))
89
90
91 # -----

```

11 Discussion and Conclusion

11.1 What the framework delivers

This paper assembles four pieces of mathematical machinery that, together, suffice to operate a derivative venue on top of any verifiable probability process:

- (i) A logit-state-space SDE with an explicit, martingale-forced drift (5);
- (ii) A closed-form funding-rate identity (8) for the perpetual future, compatible with the amortized-perpetual mechanic of [Bichuch and Feinstein \[2026\]](#);

- (iii) A Gauss–Hermite logit-normal pricer for European options with explicit Greeks (21)–(23) and a CRR binomial fallback for early-exercise extensions;
- (iv) A SPAN-style portfolio margin engine augmented by the mirrored-Weibull tail add-on (32) of Jia and Lee [2026] and the Glosten–Milgrom–Nakamura microstructure correction.

11.2 Open problems and routes forward

Four research directions naturally extend the present framework.

Stochastic volatility on the logit. Replacing constant σ with a Heston-style stochastic-volatility process introduces a volatility-of-volatility parameter and breaks the closed-form Gauss–Hermite identity; the binomial scheme of Section 6 generalises to a bivariate tree.

Multi-event copula structure. A portfolio of K correlated events admits a K -dimensional logit process whose covariance can be shrunk via Ledoit–Wolf or modelled by the GNN architecture of Wade [2026]; the SPAN grid would extend to a K -dimensional scenario hypercube and the curse of dimensionality would force a shrinkage to principal-component shocks.

Liquidity-aware execution. The half-spread of Nakamura [2026] combined with the Leland [1985] rebalancing frequency provides the building blocks for a market-maker quoting policy; numerical optimisation of the joint problem is left for future work.

DCM-grade clearing software. Building a Part-38-compliant [Commodity Futures Trading Commission, 2024] clearing front-end on top of the engine described here is the next operational step; integration with the prediction-market data feeds documented in Kalshi Exchange [2024], Polymarket [2024] is a natural calibration source.

11.3 Closing

Prediction markets have, for two decades [Wolfers and Zitzewitz, 2004, Hanson, 2003, Ostrovsky, 2012], been treated as informational instruments. We argue that they are also *economic* instruments awaiting a complete derivative architecture; this paper sketches what that architecture might look like, with enough mathematical and computational detail to make the picture operationalisable.

Reproducibility. All code, figures, and tables can be regenerated with `python3 code/make_tables.py && python3`

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A Notation summary

Symbol	Meaning
$p_t \in (0, 1)$	risk-neutral probability of event A at time t
$\ell_t = \text{logit } p_t$	logit of the probability index
$\sigma > 0$	logit-volatility per $\sqrt{\text{year}}$
μ_t	logit drift, pinned by Proposition 2
$\hat{\epsilon}_t$	standardized one-period logit innovation (6)
F_t	ELPF mark price
f_t	instantaneous funding rate (8)
C_0, P_0	European ELO call, put price
$\Delta, \Gamma, \nu, \Theta$	Greeks of the European ELO
M^{SPAN}	SPAN-style portfolio margin
s	Glosten–Milgrom–Nakamura half-spread

B Why the logit?

Three alternative state-space transforms for p_t are commonly seen in the literature:

- **Identity.** $\ell_t = p_t$. Ill-posed under diffusion as discussed.
- **Probit.** $\ell_t = \Phi^{-1}(p_t)$. Lacks closed-form moments under a Brownian driver.
- **Complementary log-log.** $\ell_t = \log(-\log(1 - p_t))$. Asymmetric; rarely used outside actuarial science.

The logit is the canonical link in the exponential-family logistic regression, admits a closed-form inverse and derivative $\frac{d}{d\ell}\sigma(\ell) = \sigma(\ell)(1 - \sigma(\ell))$, and is the unique transform compatible with the binary logistic likelihood, making calibration to historical resolution data straightforward.

C Edge cases at the boundary

When $p_0 \in \{0, 1\}$ exactly, the logit blows up and the SDE (3) becomes ill-defined. In practice prediction-market prices never exactly reach the boundaries until terminal resolution, and the Gauss–Hermite pricer clips inputs to $(\epsilon, 1 - \epsilon)$ with $\epsilon = 10^{-6}$. The clipping introduces an error of order ϵ on the price and is negligible for any practical use.